1 Basics of Information and Communication Technologies

Introduction

Information and communication technology (ICT) has been passed for generations and it dates back to antiquity. The term 'information and communication technology' has several meanings. ICT is an electronic method of conveying, processing, storing, creating, displaying, sharing, or exchanging data. Many individuals and groups have voiced their opinions on ICT.

Darnton and Giacoletto's View

- "It is the systematic study of artefacts that can be used to give form or descriptions to facts in order to provide meaning or support for decision making and artificial that can be used for the organisation and application of information."
- Thus, comprehensively, ICT can be defined as the use of one of the basic building blocks of modern society; many countries now regard understanding ICT and mastering the basic skills and concepts of ICT as part of the core of education, alongside reading, writing, and numeracy.

Smith and Campbell's View

"A mosaic of technologies, products, and technologies have combined to provide new electronic dimensions to information management. The mosaic is known by the name of information and communication technology."

Organisation for Economic Cooperation and Development View

The term 'information and communication technology' is used to cover technologies used in the collection, processing, and transmission of information. It includes microelectronic and info-electronic based technologies incorporated in many product and production processes and increasingly affecting the service sector. It covers inter alias computers, electronic office equipment, telecommunication, industrial robots and computer controlled machine, electronic components and software products.

United Nations Educational, Scientific and Cultural Organisation View

"Information and Communication Technology (ICT) is scientific, technological and engineering disciplines and management techniques used in information handling and processing, their application, computer and interaction with man and machines, and associated social, economic, and cultural matters."

Statement of UNESCO's Intentions Regarding Information and Communication Technologies

The United Nations Educational, Scientific, and Cultural Organization (UNESCO) works to guarantee that all people, in all nations, have access to high-quality education and not help advance national literacy. As a practical way of assisting nations, UNESCO has already released books on ICT in education, such as a curriculum guide published in 1994.

ICT in Various Contexts

Several crucial issues concerning data management inside an organisation are discussed in a larger framework.

Management of Information

How data are collected, checked, and stored; how data are kept safe; how information is manipulated, processed, and disseminated; and how networks are designed for information sharing all fall under the umbrella of information management.

The Nature of Information

This covers topics such as the value of information, how information is controlled, the limitations of ICT, and legal considerations.

Information Systems Strategy

- This considers how ICT can be used within a business or organisation to achieve goals and objectives.

- Information and communication technology (ICT) has become very important within a very short time with the use of hardware and software for efficient management of information, that is, storage, retrieval, processing, communication, diffusion, and sharing of information for social, economic, and cultural upliftment.
- Thus, information communication and technology is one which deals with the collection, use, storage, retrieval, transmission, manipulation, and dissemination of information accurately as efficiently as possible for the purpose of enriching the knowledge and developing communication.
- Before proceeding to the next section, let us first define the meaning of information and communication.
- Information: information is the data that is
  - Accurate and timely represented
  - Specific and organised for a purpose
  - Presented within a context and gives it meaning and relevance
  - Leads to an increment in understanding and a reduction in uncertainty
- Information is valuable because it can affect behaviours, decisions, or outcomes.
- Communication: it is defined as a process of imparting/exchanging information by speaking, writing, or some other medium. No organisation or no group can exist without information. A minimum of two people are required for the communication process. For a communication process, the requirement is
  - Source of communication or sender
  - Contents of communication or message
  - Media and the channel of communication
  - Receiver of communication
  - Feedback

ICT Tools

- Synchronous and asynchronous communication tools are used to facilitate communication between two end users.
- Synchronous communication: it occurs in real time and can occur face to face, irrespective of distance. The most commonly used synchronous tools include Skype, Google, Talk, MSN, and Yahoo Messenger. Synchronous communication gives advantages such as
  - Real time interaction
  - Immediate response and feedback
  - Cost effective solution
  - Increased social presence
- Synchronous tools suffer some disadvantages, such as
  - Difficult for one to many
  - If technology fails, not possible to interact
- Asynchronous communication: this type of communication is not immediately received or responded to between end users. Most commonly used asynchronous tools include emails, blogs, Wiki, RSS feeds, YouTube, and newsgroups. It gives the advantages such as
  - They are available at any time, any place.
  - They can provide one-to-one or one-to-many communication.
  - They can incorporate a variety of media.
- It suffers from some disadvantages such as
  - No immediate feedback
  - Written ideas can be misinterpreted
  - Lacks real time interaction

Objectives of ICT

- Formulating information system: an important objective of ICT is to formulate the information system for scientific collection, storage, and dissemination of information such that users can easily access scientific facts.
- Interpretation of information: to help the persons correctly understand and interpret the information in its true and original form is another important objective.
- Information processing: to assist the users in the information processing manually or with the help of computers.
- Using information: to help the learners (users) make effective use of information to bring desirable changes in their behaviour and develop their decision-making and problem-solving abilities.

Advantages of ICT

- Owing to the advancements in IT, it is now easier to reach more people at a lower cost

and in less time. Texting, web conferencing, and emailing people all over the world have made it possible to have immediate two-way conversations from the convenience of one's pocket.

- In the Internet age, the world has truly become a small community. Globalisation has not only helped bring people from all over the globe united, but it has also made the economies of the world into a unified, interconnected whole. As a result, we are able to transmit information and ideas more quickly and effectively. Along with this, we are also able to break through barriers of language and geography.
- There has been a decrease in discrimination and an increase in understanding because of advances in technology, which have let individuals of various nationalities interact with each other and share their perspectives.
- Efficiency in terms of monetary output: with the aid of IT, businesses have been able to computerise their operations, making them more efficient and thus more profitable. Consequently, productivity rose, bringing in a profit that allowed for higher wages and more comfortable workspace cultures.
- The generation of new employment opportunities, including, but not limited to, positions for coders, analysts, developers, and UI/UX designers.
- Since the advent of the Internet, businesses may now serve customers around the clock, seven days a week. As a result, customers from all over the world may shop at their convenience, regardless of time or location. As a bonus, one would have to leave the house to get their purchases.

Disadvantages of ICT

- The uncertainty of someone's employment situation, given the rapid pace at which the technology is evolving.
- Although IT has helped businesses run more efficiently, it has also resulted in layoffs and downsizing. This implies that many entry-level and middle management positions have been eliminated, leading to a rise in the unemployment rate.
- Consumers are concerned that their personally identifiable information may become publicly known due to hacking of mobile phone network interceptions and mail systems. internet based communication channels may be more time and effort efficient, but they also introduce new security concerns.
- The predominance of English compared to alternative languages, as seen by the prevalence of the English language on the web, is a dominant factor.
- The material of the web is corrupting today's youth, creating an ethical conundrum directly attributable to the rise of IT.
- Citizens in developing countries such as India, Malaysia, or underdeveloped countries are increasingly influenced by the culture of developed ones.

Role of ICT in Day-to-Day Life

- ICT includes the following: telephones, electronic media, various methods of voice only and video processing with the transmission, along with network based administrative control mechanisms. All the various bits of hardware and software that facilitate the exchange of data and ideas are together referred to as ICT. Everything from servers to cables to programs is included. ICTs are employed in many different aspects of everyday service delivery.
- A wide range of activities that take place through the World Wide Web, such as online shopping, banking, and even government, comes under internet services.
- Customers may check their financial accounts, pay bills, buy gift cards, transfer funds to each other, and top up their mobile phones all from the comfort of their homes with online banking, also referred to as e-banking. Consumers save their time and money and have around-the-clock access, all while having as little hassle as possible.
- Electronic commerce or e-commerce is a method of doing business that facilitates the

viewing and buying goods and services through the Internet.

- The term e-governance refers to the implementation of digital systems to enhance civilian access to state data fulfilling a fundamental human right.
- Online learning refers to a method of teaching and learning that makes use of ICTs but does not include face to face interaction between teachers and students. It refers to any kind of education or training that makes use of electronic means of communication.
- Textbooks only exist in the form of textual or graphical representation on paper, whereas e-learning materials contain audiovisual capabilities along with the ability to virtually observe classes live which are two major advantages of online based learning or e-learning. One of the best examples is students attending the class, which is taking place at some other place with the help of VSAT technology.
- E-learning is advantageous to the student as
  - Adaptive time place and pace of learning
  - Access and availability of education
  - Unlimited repetition of lectures
  - Reduces cost of education
  - Multimedia environment (video, audio, text)
- E-learning is beneficial to teachers and educational institutions as
  - It reduces the cost of teaching (as one can access it online, it reduces classroom costs).
  - Once created, it is easy to upgrade.
  - E-education allows academic staff to invest a much shorter time commitment to implement the respective education.
- Tele networking: it includes a public switched telephone network (PSTN). It operates based on circuit-switched by national, regional, or local telephony operators. Various factors such as quality of IT infrastructure, good computer literacy, etc., affect the performance of tele networking. Tele networking is advantageous as
  - It reduces costs and employee transportation reimbursement.
  - It reduces travelling time from one place to another.
  - It allows the possibility of working in a convenient place.

Email, IM, VOIP, Blog, PODCAST

- Email is the transmission of text messages using the internet. With the help of the internet, users can exchange files such as images, multimedia, documents, etc. The email consists of a user name and a domain name.
- Instant messaging (IM) service is used for instant message exchange and real time communication. It allows video conferencing also. The most popular IM services are Windows Live Messenger, Skype, etc.
- Voice over internet protocol (VOIP) is a technology that allows digitalisation and transfer of voice over the internet. It enables sound communication, such as telephone calls over the internet.
- A blog is a form of internet media where one can post or write their opinion on any topic. It can be personal or in the form of journals. Blogs are popular because end users do not need to have advanced computer knowledge, knowledge of webpage creation, etc. Blogging allows the exchange of interests, ideas, opinions, etc.
- POD-Personal on Demand + Broadcast (PODCAST) is digital containing audio or video content subscribed to and downloaded automatically through web syndication or streamed online to a computer or mobile device.
- Social networks such as Facebook, Twitter, forums, IM services, and blogs are all examples of virtual communities. When a large number of individuals have the same passions and communicate online using tools such as chat rooms and message boards, they are said to have formed a virtual community, even if they have never met in person.
- Similar to bulletin boards, where users post messages and others respond with more text, forums are online communities where people may discuss various topics. For example, TED's talk based format.

Practice Exercise

1. Consider the following impact of ICT in daily life.

I. Provides faster communication
II. Easy information sharing
III. Reduces paperwork

Select the correct answer using the codes given below.

(A) I and II only
(B) II and III only
(C) I, II, and III
(D) I and III only

2. Consider the following statements related to ICT.

I. It includes hardware and software.
II. Distance education is possible with the help of ICT.
III. It helps in sharing resources, etc., between various institutes.
IV. It provides a scalable solution.

Select the correct answer using the codes given below.

(A) I and II only
(B) II and III only
(C) I, II, and III only
(D) All of the above

Answer Key

1. (C) **2.** (D)